
PROJECT TITLE: GLAROUTES

Programme: MCA

Project Group Members:

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Objective: The GLARoutes App is a specialized Android-based navigation and assistance system developed exclusively for GLA University to simplify on-campus movement and improve accessibility. The app provides a real-time map of the campus, enabling users to locate academic blocks with accuracy and ease.

Tools required:

➤ **Hardware Requirements:**

- Laptop or Desktop Computer with minimum 8 GB RAM
- Intel i5 Processor or higher
- Minimum 256 GB SSD / HDD storage
- Android Smartphone (Android 8.0 or above) for testing
- Stable Internet connection for API integration and Firebase access

➤ **Software Requirements:**

- Android **Studio** (latest stable version)
- Java **Development Kit (JDK 8 or above)**
- Firebase **Console** (for Authentication & Database)
- Google **Maps API** (for navigation and location tracking)
- Operating **System:** Windows 10 / 11 or Ubuntu 20.04+
- Emulator for testing

Abstract: GLARoutes is an innovative Android-based navigation application designed specifically for GLA University to enhance on-campus mobility and accessibility. Unlike generic navigation tools such as Google Maps or the university's web-based map, which fail to provide precise positioning aligned with the campus infrastructure, GLARoutes offers a real-time experience by displaying the user's current location within the university premises. Developed using Android Studio, the app enables students, faculty, and visitors to navigate the university effortlessly, identify their exact position, and locate academic blocks with accuracy.

During development, extensive research and exploration of multiple campus buildings were conducted to ensure reliable mapping and detailed coverage. The app bridges the gap between global navigation systems and localized campus requirements, making it user-friendly and context-aware. By providing an accurate and interactive navigation experience tailored to GLA University, GLARoutes aims to improve convenience, save time, and enhance the overall campus experience.

Outcome:

